

Technology Stack

Date	2 JULY 2026
Team ID	xxxxxx
Project Name	Flood Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML, CSS, Bootstrap, JavaScript</i>	<i>Provides a responsive, user-friendly interface for entering weather data and displaying flood prediction results.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	<i>Flask is lightweight and integrates easily with Machine Learning models for fast prediction processing.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset / SQLite (or MySQL if you used it)	<i>Stores historical weather data and prediction records efficiently for model training and analysis.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	IBM Cloud	<i>Enables scalable deployment, remote accessibility, and reliable hosting of the Flood Prediction application.</i>
5	Version Control & CI/CD	Git & GitHub	<i>Tracks project changes, supports collaboration, and manages source code efficiently</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib</i>	<i>Used for data preprocessing, machine learning model development, prediction, and data visualization.</i>